Proactive Camera Attribute Control Using Bayesian Optimization for Illumination-Resilient Visual Navigation

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***Abstract*—Illumination variance is a major challenge for vision-based robotics. Most approaches focus on alleviating illumination changes in already captured images. Despite the large utility, camera attributes have been empirically determined to function in a highly passive manner, yielding vision algorithm failure under radical illumination variance. Recent studies have proposed exposure and gain control schemes that could maximize image information and eschew saturation. In this article, we propose a proactive control scheme for the camera's two dominant attributes—exposure time and gain control. Unlike existing approaches, we formulate this camera attribute control as an optimization problem in which the underlying function is not known *a priori*. We first define a new metric of the image regarding these two major attributes to include both image gradients and signal-to-noise ratio simultaneously. Based on this metric, we introduce a new formulation for this attribute control via Bayesian optimization (BO) and learn the environmental change from the captured image. During the control, to mitigate the burden of image acquisition and Bayesian optimization, images are synthesized using a camera response function and avoided the actual frame grab from the camera. The proposed method was validated in light-flickering indoor, outdoor near sunset, and indoor–outdoor transient environments where light changes rapidly, supporting 20–40 Hz frame rates.**

***Index Terms*—Camera attribute control, computer vision for other robotic application, field robots, visual-based navigation.**

I. Introduction

CAMERAS are essential sensors for computer vision and robotics researches. However, their vulnerability to illumination change has been an issue for decades. Despite well-perceived concerns about illumination changes, most vision-based approaches assume little change in brightness, operating cameras in either fixed or automatic exposure mode. Unfortunately, this assumption readily breaks in a real-world scenario. For example, a sudden transition into a space with a large difference in brightness may cause failure in vision-based navigation. Recently, Yang *et al.* [2] indicated three challenges for monocular visual odometry in their article—namely, photometric calibration, motion bias, and rolling shutter effect.

Amendment to the captured image has been proposed in various aspects. Maddern *et al.* [3] newly defined an illumination invariant channel where feature matching can be performed regardless of illumination. Engel *et al.* [4] incorporated the photometric calibration term in the simultaneous localization and mapping (SLAM) algorithm to involve illumination variance within SLAM module. These methods, however, may be rather passive without taking action at the image acquisition phase. Some efforts exist in the robotics and computer vision fields to actively control camera attributes and secure better image quality in the acquisition phase. In [5] and [6], the authors introduced a metric to evaluate a captured image and a simple but effective control architecture to adjust the exposure and gain. In their works, both authors dominantly controlled the exposure first and then applied fine gain adjustment, if needed.

Sharing similar proactive philosophy, our previous method [1] provided an effective frame acquisition but only considering the exposure time. In this article, we extend our previous work to general camera attribute control by jointly considering gain and exposure time in the metric evaluation and control phase. In our previous work, the bottleneck was at the Bayesian evaluation phase when the function evaluation corresponds to an actual frame grab. By introducing camera response function (CRF)-based image synthesis, we substantially improve the evaluation time. Most importantly, our keen interest was at visual navigation performance improvement in terms of both feature-based and direct SLAM. As will be shown, our key finding proves that the proposed attribute control enables resilient perception even under several and radical illumination change including a transient between indoor and outdoor.

II. Related Works

Overcoming the high dynamic range (HDR) environment has been widely studied from the perspectives of: 1) image fusion in computer vision; 2) photometric inconsistency consideration

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Code will be available at https://github.com/irapkaist/cam_attr_contoller.

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in visual navigation; and 3) camera exposure control in machine vision. We briefly summarize the related studies in these fields.

A. Image Fusion in HDR

Under extreme illumination changes, instead of selecting one optimal exposure, multiple exposure images with different exposure values can be used to solve the exposure problem [7] and [8]. To reconstruct the HDR image suitable for human vision, Debevec and Malik [9] created a computed radiance image by defining the relationship between CRF and pixel radiance through linear mapping. In [10], the study proposed introducing a camera's radiometric response function (RF) into a single gray scale image to avoid using a set of registered images. They used the statistical properties of the gray scale histogram in the edge region to obtain information about the radiometric calibration. The resulting image covers a large variance in illumination, yet, this approach requires multiple images and is hardly applicable to real-time application.

B. Camera Attribute Considered Visual Navigation

Some researchers solved the illumination variance issue by incorporating the attribute of the captured image into the visual navigation phase. Bergmann *et al.* [11] proposed a method for online photometric calibration. The algorithm recovers the exposure times of consecutive frames, the CRF, and the attenuation factors of the sensor irradiance due to vignetting. Recently, Yang *et al.* [2] examined and evaluated the advantage and limitations of the photometric calibration over a set of SLAM methods. According to their paper, the photometric calibration significantly improved the direct (direct sparse odometry (DSO) [4]), while reducing performance over semidirect (semi-direct visual odometry (SVO) [12]) and feature-based SLAM (oriented fast and rotated BRIEF (ORB) SLAM [13]). Interestingly, photometric calibrations may reduce contrast of the dark pixels due to the slope differences in the CRF fitting phase.

C. Camera Attribute Control

Others tackled the problem by controlling camera attributes—namely, exposure time and gain, and presenting image quality metrics and control schemes.

In early works, image intensity was used to distinguish the light or dark parts of the image [14]. This ensures that the region of interest (ROI) is accurately exposed [15], [16]. This ROI-based method is limited in certain known environments and a more recent approach relies on the level of the image gradient in the control scheme. For example, the sum of gradient magnitude [5] or the percentile of each pixel gradient [6] were introduced. Image entropy is another decisive factor. Exposure mapping using the image with the highest entropy information is suitable for robot vision [17] because it shows the level of diversity within the image. However, since image entropy is based on a probability model, it is difficult to predict the exposures in an HDR environment, so camera parameters can only be adjusted using empirical methods.

Shim *et al.* [5] generated several individual synthetic images using the gamma function, found an image with the largest gradient value among the generated images, and then mapped it at the corresponding exposures. However, the actual image obtained from the camera and the image generated using the gamma function may be different. Zhang *et al.* [6] used the photometric RF and the image gradient to find optimal exposures. The derivative of the squared gradient was calculated and then used to estimate the sum of gradient magnitude over exposure time through the photometric RF. Another method [17] is to compute the image entropy, measure the amount of information in the image, and then determine the gain and exposure time accordingly. They select image parameters with maximum entropy and does not consider the control scheme.

Automatic gain control is one of the functions that control camera exposure, which increases the gain when the image is too dark and reduces the gain when the image is too bright [18]. Gain increase and decrease between adjacent frames are limited to 1 dB so that the gain does not oscillate. Litvinov and Schechner [19] introduced a method of simultaneously estimating the nonlinear RF and camera gain in the image sequence using a radiometric RF.

III. Overview

A. Problem Definition and Contributions

Among the three main attributes that determine image exposure, (i.e., gain, aperture, and exposure time), we focus on the two attributes of exposure time and gain, and examine their relationship in the attribute control. Given a seed image (I_o) and incoming image frame I, the algorithm seeks for the optimal camera attribute pair, exposure time and gain ($\Delta t^*, G^*$). When determining the optimal camera attributes, we introduce a generic and proactive camera attribute controller that simultaneously controls the exposure time and gain. In summary, the proposed method is novel in presenting the following contributions.

1) We propose a novel metric for image information measure by considering gradient, entropy, and signal-to-noise ratio (SNR). The level of noise and gradient counterbalance each other to avoid extreme saturation or sensitivity.
2) We provide a unified approach to actively control camera attributes, (i.e., gain and exposure time). These two major factors are often determined independently or sequentially in the literature. In this work, we analyze the impact of exposure time and gain simultaneously via experiments in various lighting environments.
3) We formulate this attribute control as a Bayesian optimization (BO) and find optimal camera attribute values through an internal prediction step. Furthermore, to alleviate the cost of capturing a real image during evaluation, we suggest image synthesis based on CRF in the acquisition module.

B. Control Architecture

Overall control architecture is as illustrated in Fig. 1. The camera attribute controller is initiated by capturing a seed image to evaluate the environment. Using this seed image, the synthesizing module generates synthetic images for use in the BO module. The computed optimal camera attribute is assigned to capture

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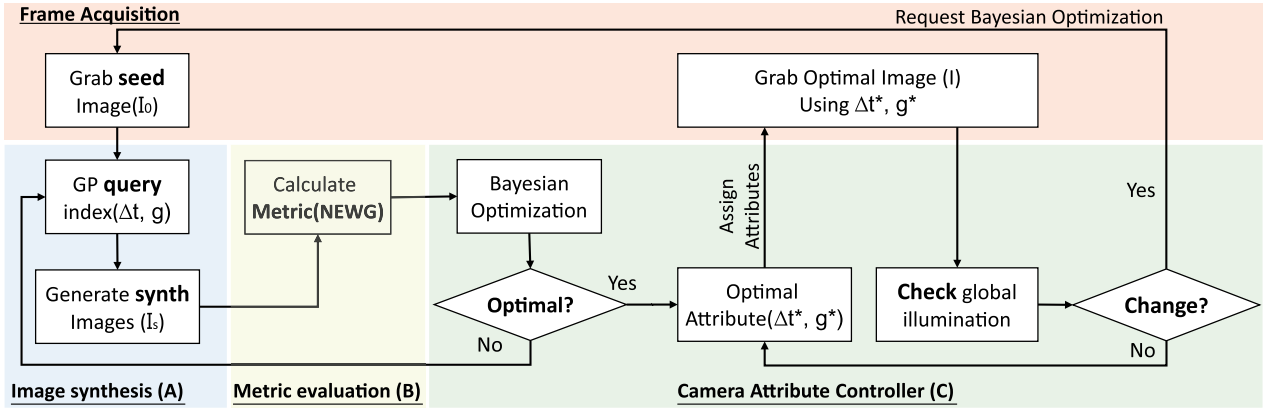


Fig. 1. Diagram for the proposed method. The overall procedure consists of three modules: 1) image synthesis (blue box, A); 2) metric evaluation (yellow box, B); and 3) control module (green box, C).

a real image until a global illumination change is detected. For every incoming image frame, we examine the occurrence of changes in illumination by comparing the image metric. Only when the illumination variance is detected, revealing a large discrepancy between synthesized image and actual frame grab in terms of the metric, the entire BO phase is requested.

The proposed method consists of three modules—image metric evaluation, control, and image synthesis module. First, we take the initial seed image, compute the gradient and entropy, and minimize the noise in the image gradient through the weights calculated from the entropy information (Section IV). Next, as a metric compensation step, local entropy of image also generates a saturation mask and compensates the entire metric with an activation function. Once the metric calculation is complete, a series of steps is repeated until a Gaussian process (GP) finds optimal attributes (Section VI), producing synthetic images (Section V) by using the CRF and gain functions for the next query image to be learned.

IV. Image Utility for Robot Vision

In this article, we extend our previously developed entropy-weighted gradient (EWG) [1] and moderate it using the noise level measure from signal-to-noise ratio (SNR). Doing so completes a unified image utility measure considering saturation (entropy), edges (gradient), and noise (SNR). The motivation behind using entropy as the measure of saturation is to reward the local patch with higher diversity while completely penalizing the saturation. Even a monotone-colored wall may have a slight variation while the saturated region shows no variation in the local image patch.

A. Entropy-Weighted Image Gradient

We summarize the EWG from our previous publication [1]. Adopting the entropy definition introduced in [20], the local image entropy for pixel i is written as

$$H_{\text{local},i} = -\sum_{k} P(i_k)\log_2(P(i_k)) \tag{1}$$

where $P(\cdot)$ is the probability associated with the number of intensity levels (k) within 5×5 local patch. We normalize this entropy over an image $H_i = H_{\text{local},i} / \sum_{i=0}^{N-1} H_{\text{local},i}$ and define the weight for an image pixel as

$$w_i = \frac{1}{\sigma}\exp\left(-\frac{(H_i - \text{mean}(H_i))^2}{2\sigma^2}\right) \tag{2}$$

while σ captures the variance. Then, we normalize this pixel-wise weight function as $W_i = w_i / \sum_{i=0}^{N-1} w_i$.

Next, we define the activation function $\pi(\cdot)$, given the computed entropy-based function as

$$\pi(H_i) = \frac{2}{1+\exp\left(-\alpha H_i + \tau\right)} - 1,\ 0 \le H_i \le 1. \tag{3}$$

We then generate a binary saturation mask (M) for pixel i as follows:

$$M_i(H_i) = \begin{cases} 0, & H_{\text{thres}} \le H_i \le 1 \\ 1, & 0 \le H_i < H_{\text{thres}} \end{cases}. \tag{4}$$

If the entropy value for a pixel i is lower than a certain threshold ($H_{\text{thres}} = 0.05$, in our implementation), then the pixel is considered saturated and the entire metric is penalized. Fig. 2 shows the effect of $\pi(\cdot)$ and mask M.

The metric and activation function are based on the finding that the local entropy is minimum for the saturated pixels. The combined metric is then written as follows:

$$u_i = W_i \underbrace{\| \nabla I(i) \|^2}_{\text{Image gradient}} \underbrace{+\pi(H_i)M_i(H_i)}_{\text{Penalty mask}} W_i \frac{1}{N}\sum_{j=0}^{N-1} \| \nabla I(j) \|^2 . \tag{5}$$

The multiplied term becomes negative and behaves as a penalty term. Combining the image gradient with the entropy-induced weight, activation function, and mask completes our entropy-weighted metric. A sample image, entropy, and entropy weighted gradient are shown in Fig. 3.

The image gradient for image I is represented by a derivative with respect to pixels and the gradient magnitude of pixel i

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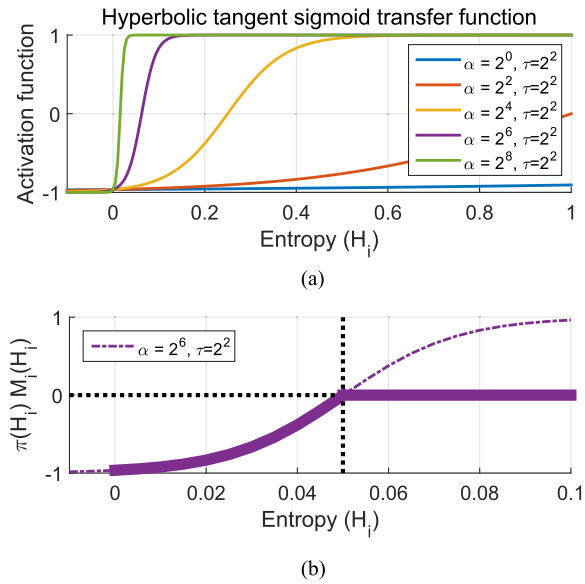


Fig. 2. (a) Activation function maps entropy of a pixel to activation level. The activation level is controlled by two parameters, α and τ. (b) Plot of $\pi(H_i)M_i(H_i)$ in the second penalty term. Negative weight is imposed to penalize the saturation.

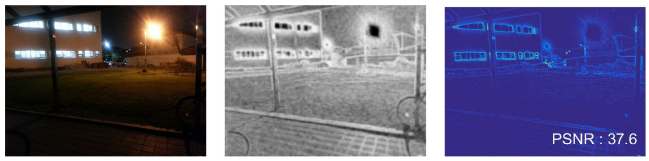


Fig. 3. Entropy and EWG of an image. The Black regions mean no change in intensity (i.e., little image information). (a) Original image. (b) Entropy. (b) EWG.

is indicated by $\| \nabla I(i) \|^2$. We consider the weight, activation function (3), binary mask (4), and image gradient when calculating the EWG (u_i) for pixel i.

The first term, EWG, is critical for robot vision applications. However, for visual navigation, this may yield a radically saturated image targeting only on the gradient. As a balancing term, we introduce the second term by using the activation function-considered penalty mask to balance between image gradient and saturation.

The final utility metric for measuring image information for exposure time is the sum of the EWG for the total number of pixels (N)

$$U_t = \sum_i u_i. \tag{6}$$

This EWG allows us to balance between gradient and saturation by rewarding gradient and penalizing saturation simultaneously.

B. Image Information Measure Using SNR

The previous utility metric focused on the oversaturation from the exceeded exposure value. In cases of low luminance environments and sudden diminishing of the light source, gain control would be preferred over increased exposure time to avoid motion blur. To reflect the increase in noise, we define a noise considered metric using SNR.

The SNR of an image is typically calculated as the ratio of the mean pixel to the standard deviation of the pixel for a given neighbor. In general, the higher the gain, the lower the SNR and the lower the image quality. The International Organization for Standardization (ISO) sets the industry standard for SNR and measures sensitivity. According to [21], 32 dB is defined as superior and 20 dB is defined as acceptable image quality. In this article, we use an acceptable $\text{SNR}_{\text{ref}} = 30$ to weight the entire image gradient. We compute the SNR metric for entire image I as

$$(U_k)_I = \left(1 - \frac{\text{SNR}}{\text{SNR}_{\text{ref}}}\right)\left(\frac{1}{N}\sum_{j=0}^{N-1} \| \nabla I(j) \|^2\right). \tag{7}$$

Given two terms for exposure time $(U_t)_I$ and gain $(U_k)_I$, we sum the two terms as in (8). We will denote this final metric, image utility, as U_I throughout the article.

For an image I, the proposed evaluation metric U_I is the sum of the metrics exposure time $(U_t)_I$ and gain $(U_k)_I$

$$U_I = (U_t)_I - \frac{1}{\kappa}(U_k)_I \tag{8}$$

where κ is the user parameter controlling the penalization balance between the exposure time and the gain. In this work, $\kappa = 5$ was used. By associating the noise term with EWG, we introduce a noise-considered EWG (NEWG), which will be used in the following control scheme.

V. Synthetic Image Generation

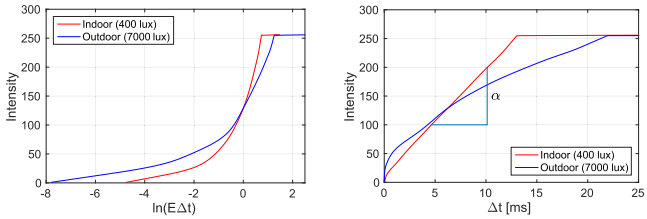
To formulate a Bayesian optimization (BO), the proposed control scheme requires function evaluation to find the optimal attributes. The issue arises when this evaluation is an actual frame grab corresponding to a certain attribute pair. Frame acquisition itself may be fast, but imposing a larger exposure time would cause an unfavorable time delay in each acquisition. To alleviate this issue and enhance computational efficiency, we propose CRF-based image synthesis to generate a synthetic image for the requested attribute pair. Specifically, we create and use synthetic images corresponding to the target exposure times and gain values avoiding actual camera frame grab. For synthesizing, we use camera attribute pair ([0 dB, 1 ms] for indoor) and ([0 dB, 50 μs] for outdoor) during the seed image acquisition.

Our method is based on CRF and generates a synthetic image for a target attribute pair by multiplying the scaling factor K_t for exposure time and the scaling factor K_g for gain to a seed image. This seed image is captured from the camera and downsampled at the initial phase with minimum exposure time and when required from the controller.

A. Image Synthesis Using CRF

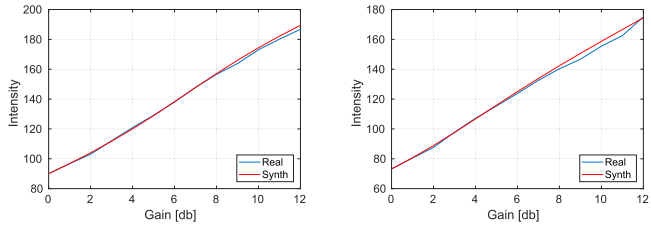
We begin by briefly reviewing CRF in machine vision research. CRF represents the relationship between a camera's irradiance and the measured intensity. Research to estimate CRF has

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(a) (b)

Fig. 4. Estimated log of inverse CRF. The intensity is evaluated over randomly sampled pixel points. (a) Estimated log of the inverse CRF. (b) Synthesizing scale.



(a) (b)

Fig. 5. Average image intensity of the real and synthetic image are plotted while changing the gain. Images are synthesized via (14) by using the image intensity captured in 0 dB as the seed image (I_o). (a) Indoor (400 lux). (b) Outdoor (7000 lux).

been widely conducted in the field of computer vision [9], [10], and [22]. Assuming CRF is spatially uniform within the image, it can be estimated extensively through different sets of exposures with the same scene. The nonlinear relationship of CRF expressed by [9] is defined on a pixel level, however, the purpose of this procedure is not to create an ideal image but to quickly and accurately create a composite image of the exposure to find the optimal camera property. We start with the CRF in [9] that defines the relationship between irradiance E_i for pixel i when captured with exposure time Δt_j as

$$I_{i,j} = f(E_i \Delta t_j). \tag{9}$$

Although CRF is unknown, we already know that the function is continuous and monotonic. Many High Dynamic Range (HDR) imaging researchers use this function property to focus on the inverse of CRF [23]. Similarly, we introduce the log of inverse CRF as $g(\cdot) = \ln f^{-1}(\cdot)$.

Given an image and exposure time pair, the authors in [9] estimated $g(I_{i,j})$ and E_i by solving a least square problem for randomly sampled pixel points

$$g(I_{i,j}) = \ln(E_i \Delta t_j) = \ln E_i + \ln \Delta t_j. \tag{10}$$

We mainly follow the method introduced in [9] by excluding saturated pixels. According to [9], at least two image acquisitions with sufficiently different radiance values are needed to estimate $g(I_{i,j})$ and irradiance E_i. In this work, we fit the log of the inverse CRF using two sample points at the boundary. Minimum and maximum exposure times are evaluated to generate two samples for the fitting. Fig. 4(a) presents an estimated log of inverse CRF $g(I)$ for two different irradiance values (i.e., indoor and outdoor). This plot is depicted using $I = g^{-1}(\ln E \Delta t)$, and the shape changes as the irradiance changes.

Unlike the previous method of matching RF exactly through the image set, we focus on leveraging this RF for synthetic images for a certain exposure time. In [23], the authors computed a synthetic image at a target exposure time (Δt_2) using an obtained image at the current exposure time (Δt_1), inverse CRF, and the ratio between Δt_1 and Δt_2. Motivated by [23], we searched for the proportional relationship between intensity and exposure time for image synthesis. In doing so, we only focused on the average irradiance by averaging over the estimated pixel-wise irradiance. By examining the obtained $g(I)$ in Fig. 4(a), we found that removing log and dividing by the average irradiance E reveals a locally linear characteristic of the relationship between intensity and exposure time as shown in Fig. 4(b). More specifically, we can rewrite (10) by taking the exponential and dividing with the computed E over an image. Assuming constant irradiance (E) of the image, we use two real obtained images (I_1 at Δt_1 and I_2 at Δt_2) to reckon $h(\cdot)$ as follows, which corresponds to Fig. 4(b):

$$\frac{1}{E} \exp\left(g(I)\right) = \Delta t = h(I) \text{ and } I = h^{-1}(\Delta t). \tag{11}$$

Although we can use $g(\cdot)$ to compute the scaling factor, we found that precomputing and storing ratio α from locally linear behavior is more practical. This ratio α can be precomputed via $h(\cdot)$ as depicted in Fig. 4(b)

$$\alpha = \frac{h^{-1}(\Delta t_2) - h^{-1}(\Delta t_1)}{\Delta t_2 - \Delta t_1}. \tag{12}$$

When we need to synthesize images corresponding to Δt_s, this precomputed ratio can be used to obtain the scaling factor as

$$K_t = \frac{I_s}{I_o} = \frac{\alpha(\Delta t_s - \Delta t_o) + I_o}{I_o} \tag{13}$$

where I_s is the average intensity of the synthesized image.

B. Image Synthesizing for Gain

Determining scale factor K_g for gain is straightforward using the definition for gain $g = 20 \log_{10} \frac{I_s}{I_o}$. Because we wish to synthesize an image with gain factor K_g, the above equation can be further simplified as

$$I_s = K_g I_o = (10/F_n)^{\frac{g}{20}} I_o \simeq 7.01^{\frac{g}{20}} I_o. \tag{14}$$

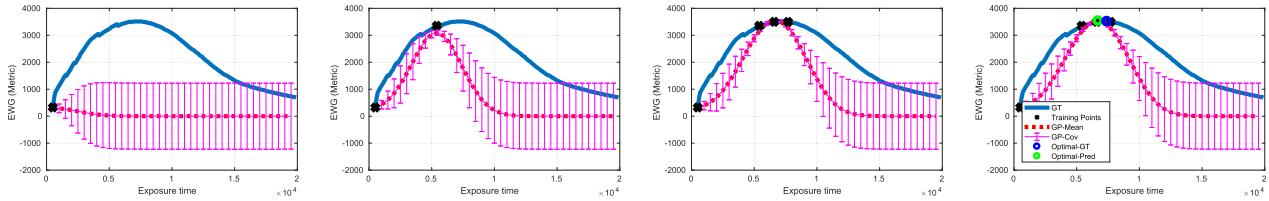
Factoring with $F_n = \sqrt{2}$ was empirically determined. For indoor and outdoor evaluation datasets, we plotted the average intensity of both real and synthetic images by varying gain as in Fig. 5. As can be seen in the figure, the scaling factor successfully captures the average image intensity for both indoor and outdoor settings.

A scale factor for exposure time (K_t) and gain (K_g) is derived to finally generate the synthetic image that follows:

$$I_s = K_t K_g I_o. \tag{15}$$

These synthesized images will then be used in the function evaluation step substantially reducing overall control time.

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(a) (b) (c) (d)

Fig. 6. Procedure of GP-based exposure control scheme. Only 1-D (Δt) illustration on exposure time was plotted for better visualization. Note that the entire scheme was performed in a 2-D (Δt, G) search space. Blue lines represent the ground truth metric (score) function and red lines indicate estimated mean of GP inference. Black dots are training points [EWG in (6) of captured image]. Figures indicate querying procedures. Subscript i in t_i is the number of querying step. Figure (d) shows the final inference mean and selected optimal exposure (green circle). (a) t_0. (b) t_1 step. (c) t_4 step. (d) Find optimal value (t_5).

The resulting images from the synthesis are provided in Fig. 11 in comparison to real images.

VI. Exposure Control

This section describes the BO-based camera attributes control schemes. We formulated camera attributes control as an optimization problem although the underlying function is not known but varies with respect to exposure time and gain. In solving this problem, we found that GP optimization [24] allows us to search for optimal camera attributes with minimal querying evaluation. Under radical illumination change, we restart the BO to avoid camera control drift and to maintain real-time performance.

A. Bayesian Optimization

With the pretrained hyperparameters, we constructed a camera controller as Bayesian inference. The expression of the predictive distribution is

$$\mu(x_*) = \mathbf{k}_*^T (K + \sigma_n^2 I)^{-1} \mathbf{y} \tag{16}$$

$$\sigma^2(x_*) = k(x_*, x_*) - \mathbf{k}_*^T (K + \sigma_n^2 I)^{-1} \mathbf{k}_* \tag{17}$$

where μ and σ indicate mean and covariance of the estimated image metric. Inputs $\bar{\mathbf{x}} = (x_0, \ldots, x_m)$ and $\bar{\mathbf{y}} = (y_0, \ldots, y_m)$ are training points that represent the camera attributes $\mathbf{x} = [\Delta t, G]$ and the related image utility measure U_I in (8). Kernel matrices $K = [k(x_i), k(x_j)]_{x_i, x_j \in \mathbf{x}}$, and $\mathbf{k}_* = [k(x_i, x^*)]_{x_i \in \mathbf{x}}$ are constructed by the kernel function $k(\cdot, \cdot)$. We selected squared exponential for the kernel function. When applying GP optimization to the camera attribute control problem, two factors are critical in terms of computational speed—choice of query and terminating criteria.

Query selection: For more efficient camera control, we adopted a query selection policy based on maximum mutual information when selecting the sampling inputs (i.e., camera attribute pair and image metric) [25]. In the main Algorithm 1, instead of random or linear selection, the sampling inputs are determined incrementally through the query selection algorithm. The query selection searches for the balanced query point between exploitation and exploration by maximizing the summation of the mean (exploitation) and the cost (exploration) terms as follows.

$$x_n = \underset{x}{\operatorname{argmax}}\, \mu_t + \Phi_t. \tag{18}$$

Algorithm 1: Exposure Time and Gain Control.

Input: Initial Train Data $\mathbf{x}_0$ and $\mathbf{y}_0$, Query Exposures $\mathbf{x}_q$
$\eta_0 \leftarrow 0$
while $!isOptimal$ **do**
  $\mu_t, \Sigma_t \leftarrow$ GP-Inference$(\mathbf{x}_{t-1}, \mathbf{y}_{t-1})$
  $x_n, \eta_t \leftarrow$ SelectQuery$(\mu, \Sigma, \eta_{t-1})$ ▷ Alg. 2
  $I_n, y_n \leftarrow$ GrabAndMetric(x_n)
  if CheckOptimal$(x_n, \sigma_{t-1}(x_n))$ **then**
    $isOptimal \leftarrow$ true
    $x_o \leftarrow \underset{x}{\operatorname{argmax}}\, \mu_t$
  else
    $\mathbf{x}_t, \mathbf{y}_t \leftarrow$ UpdateTrain(x_n, y_n)
  end if
end while
Output: Optimal exposure time and gain x_o

Algorithm 2: Select Query.

Input: Estimated mean and covariance μ, Σ, previous exploration parameter η_{t-1}
$\boldsymbol{\sigma}_{t-1}^2 \leftarrow \operatorname{diag}(\Sigma)$
$\Phi_t \leftarrow \sqrt{\gamma}(\sqrt{\sigma_{t-1}^2 + \eta_{t-1}} - \sqrt{\eta_{t-1}})$
$x_n \leftarrow \underset{x}{\operatorname{argmax}}\, \mu_t + \Phi_t$
$\eta_t \leftarrow \eta_{t-1} + \sigma_{t-1}^2(x_n)$
Output: Next query exposures x_n, exploration parameter η_t

Following [25], we defined the cost as

$$\Phi_t = \sqrt{\gamma}(\sqrt{\sigma_{t-1}^2 + \eta_{t-1}} - \sqrt{\eta_{t-1}}) \tag{19}$$

using inferred covariance and exploration controlling term η. The detailed algorithm is listed as 2. The optimal attributes are selected as the values that represent the maximum estimated image metric of the final GP inference. The parameter $\gamma = \log(2/\delta)$ controls the balance between exploitation and exploration. In [25], values ranging from $0 < \delta < 1$ were recommended; we used $\delta = 0.25$ to place more weight on exploitation for fast convergence. The detailed step-by-step view of optimization is shown in Fig. 6. The entire querying procedure terminates in five steps with 7300 μs.

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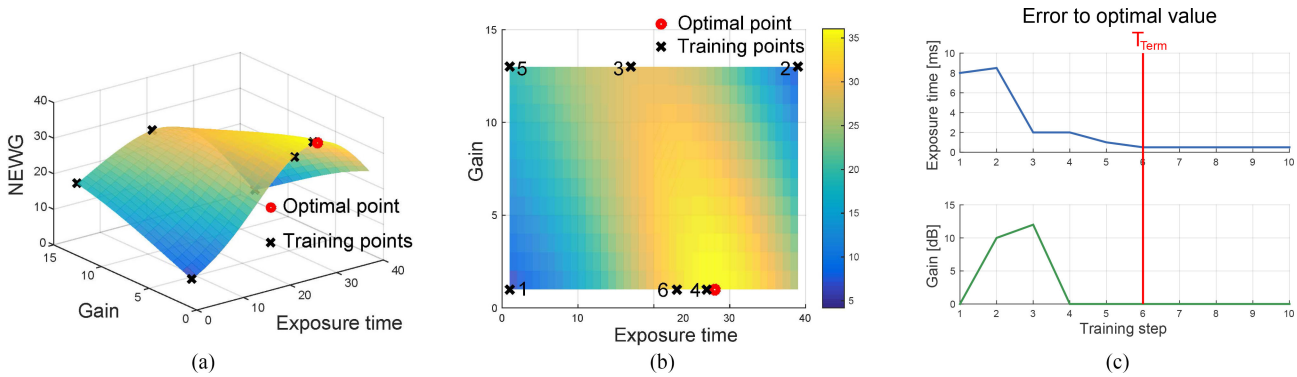


Fig. 7. Example of camera attribute control. (a) and (b) Estimated distribution using six training points (black cross) and predicted optimal values (red dot). (c) Error between the t-step's query point and the optimal value.

Terminating criteria: Another key factor for real-time control performance is the terminating criteria of the GP. The underlying function is convex and the cost converges fast, enabling us to yield the final camera attributes. The optimization process is easily extended to multiple attributes' controls. Fig. 7 represents joint optimization of camera exposure time and gain control. Using the same process with extended input dimensions (exposure and gain), the optimal attributes (red dot) can be estimated through selected inputs (black cross). Termination of the query request is determined empirically as in Fig. 7(c). Using the exhaustive dataset (Section VII-A2), we examined errors between tth query value and the optimal value. Based on this plot, we terminate the query request when the iteration exceeds $T_{\text{term}} = 6$.

B. Hyperparameters and Complexity

We also examine the hyperparameters for the kernel function, which is expressed as

$$k_{SE}(x, x') = \sigma_s \exp\left(-\frac{(x - x')^2}{2\sigma_l^2}\right). \tag{20}$$

To acquire proper hyperparameters, we verified the hyperparameters on various scenes. We captured 18 scenes of indoor and outdoor environments with varying gain and exposure values. Having training data (exposure, gain, and relative metrics), we applied log-likelihood optimization to hyperparameter selection. The optimized hyperparameters are similar because the dimension scales of the training data are fixed and metric results are smooth as the camera attributes change. The average of optimized hyperparameters was selected as the camera attribute controller, $\sigma_l = 9.74, \sigma_s = 10.42$.

Similar to other GP-based approaches, the proposed method may suffer if a large number of points are evaluated. For M number of query points, the complexity scales as $O(M^3)$. We could further reduce the time needed by providing four boundary query points in advance and using precomputed covariance. However, judging from the terminating criteria and dimensions of the query points, the computational complexity does not critically deteriorate the overall control performance.

C. Camera Attribute Assignment

When implemented, however, assigning the optimal camera attribute may yield an unnecessary flickering effect. We examine the ratio of the metric of the previous real image to the metric of the current synthetic image, and request BO only when the discrepancy is larger than 20% (i.e., $|(U_{I_{syn,t}} - U_{I_{real,t-1}})|/U_{I_{real,t-1}} > 0.2$).

VII. Experiments

To validate the proposed control scheme, we conducted a set of experiments in three aspects. First, we validated the performance of the proposed image utility while comparing it to other metrics used in existing camera control approaches. Next, we evaluated the CRF-based synthetic images against real images. Lastly, we verified the overall control scheme in a visual SLAM framework.

A. Image Utility Evaluation

1) Effect of SNR in Various Illumination: We collected data under various lighting conditions as listed in Table I. As can be seen from this table, the metric that takes into account for SNR had a better penalty for high gain values. For example, in the office environment (320 lux), low exposure time (2.5 ms) and high gain (11 dB) were chosen if noise compensation was not considered because even though the noise increases with a higher gain, more gradients are obtained. On the other hand, the noise-considered metric mitigated this problem by selecting a larger exposure time (7.0 ms) and a lower gain (1 dB). Similarly, we could obtain more balanced parameters that minimized blur and noise even in a dark environment (10 lux).

Fig. 8 compares two different metrics while sequentially changing exposure time and gain. Compared to the EWG metric, the Noise-considered Entropy-Weighted Gradient (NEWG) metric efficiently complemented the metric even in dark environments. In the dark environment (10 lux), using the SNR [see Fig. 8(d)] did not drastically increase the overall curve of the graph along exposures. In addition, by including noise compensation in the metric, as the noise increased, the penalty

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TABLE I
COMPARISON OF OPTIMAL ATTRIBUTE (Δt^* AND G^*) DEPENDING ON THE CHOICE OF METRIC

Illuminance [lux]		10	50	80	100	320	400	1000
Environment		*Dark scene*	*Living room*	*Hallway*	*Overcast*	*Office*	*Sunny*	*Artificial light*
SNR-ignored (EWG)	Δt^* [ms]	17.0	12.0	5.5	5.5	2.5	2.0	2.0
	G^* [dB]	11	3	0	0	11	4	1
SNR-considered (NEWG)	Δt^* [ms]	19.5	12.5	5.5	5.0	7.0	2.0	1.5
	G^* [dB]	9	0	0	0	1	4	1

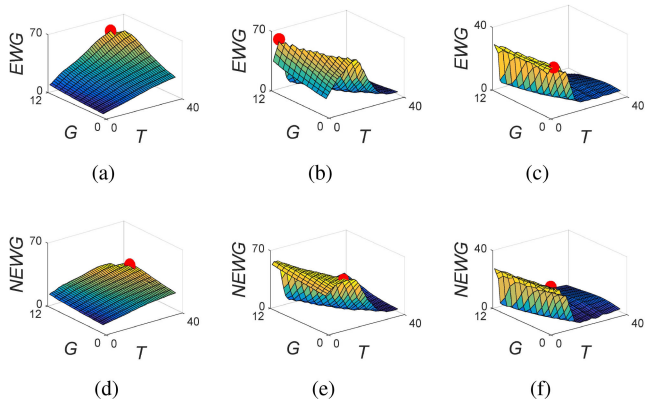


Fig. 8. Sample image evaluation metric plotted with respect to the gain and exposure time. Red dot represents the optimal value of each metric. Note the change in the optimal values for 320 lux (office) when SNR is additionally considered. (a) 10 lux. (b) 320 lux. (c) 1000 lux. (d) 10 lux. (e) 320 lux. (f) 1000 lux.

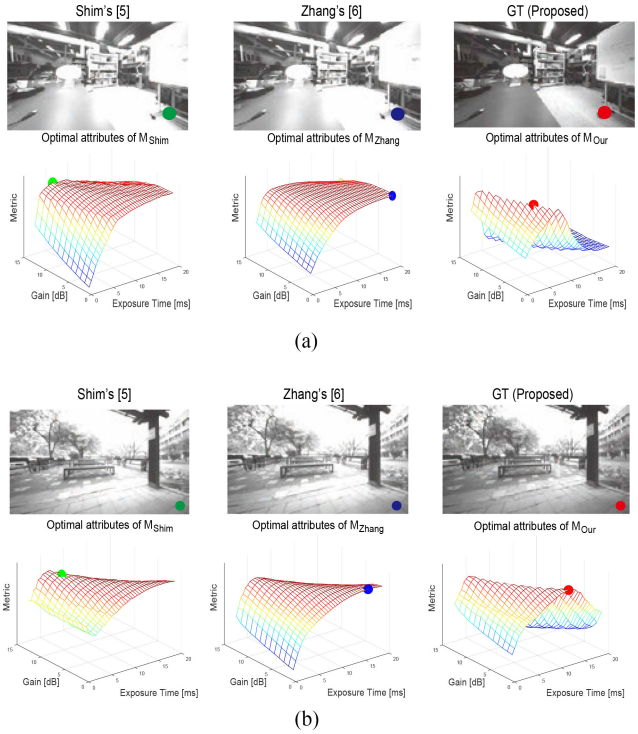


Fig. 9. Metric comparisons from (a) indoors and (b) outdoors. On the top row, a sequence of images is listed while increasing the camera exposures. Optimal exposures are selected by Shim's [5] (green), Zhang's [6] (blue), and proposed (red). Graphs depict the information metric measured by each method with respect to the exposure time and gain. Proposed metric penalizes more rapidly than other methods, preventing the camera from assigning too large of an exposure value.

determined the exposure time and gain appropriately. Ignoring SNR concentrated to eliminate saturated regions by light sources such as an office environment (320 lux). However, the SNR metric attempted to allow some saturation [see Fig. 8(e)] unless a higher noise was produced in the image. The metric also prevented the image from selecting excessive exposure time or gain, even in bright environments where the exposure time can only be controlled. Because the algorithm searches for balanced exposure times and gains, it produced results similar to previous metrics in bright environments (1000 lux).

2) Comparison to Other Metrics: Next, we compared the proposed metric against other metrics [5], [6]. For both indoor and outdoor environments, we captured a static scene while changing exposure value so as to generate a set of images under varying levels of quality. Each image quality metric plot and the associated optimal image are shown in Fig. 9. The selected optimal attributes differed depending on the underlying image quality measure. Other metrics allowed saturation in some portions to achieve more detail in the unsaturated area. Unlike others, the proposed metric compensated for the saturation region by incorporating entropy in the evaluation metric.

We compared the metric against others in terms of the feature tracking performance. While moving from the outdoors to the garage, the ORB features were tracked from images corresponding to each method. We compared the proposed method (GT) against the manual exposure control, mean intensity control [14], Shim *et al.* [5], Zhang *et al.* [6], and our previous exposure time only (T) [1]. Both the quantitative and qualitative results are shown in Fig. 10 and Table II. The effect of entropy-based saturation measure is noticeable when compared to [5], [6] that relies solely on image gradient. As shown in the T5 image, the saturation region largely occurred due to the brightness difference when entering the garage. Our method was able to reliably track features by maintaining constant brightness and minimizing saturation in such a challenging situation. In Table II, Shim's and Zhang's method selected a relatively high exposure by only focusing on more image gradient acquisition. Therefore, the saturation area of the image increased by about 20%–30%, whereas our method was less than 10%. A notable point is the image quality difference between the T and GT. For example, the image obtained from T shows a smaller saturation region and better SNR. However, the increased gain and shorter exposure time from GT enabled more gradient and features.

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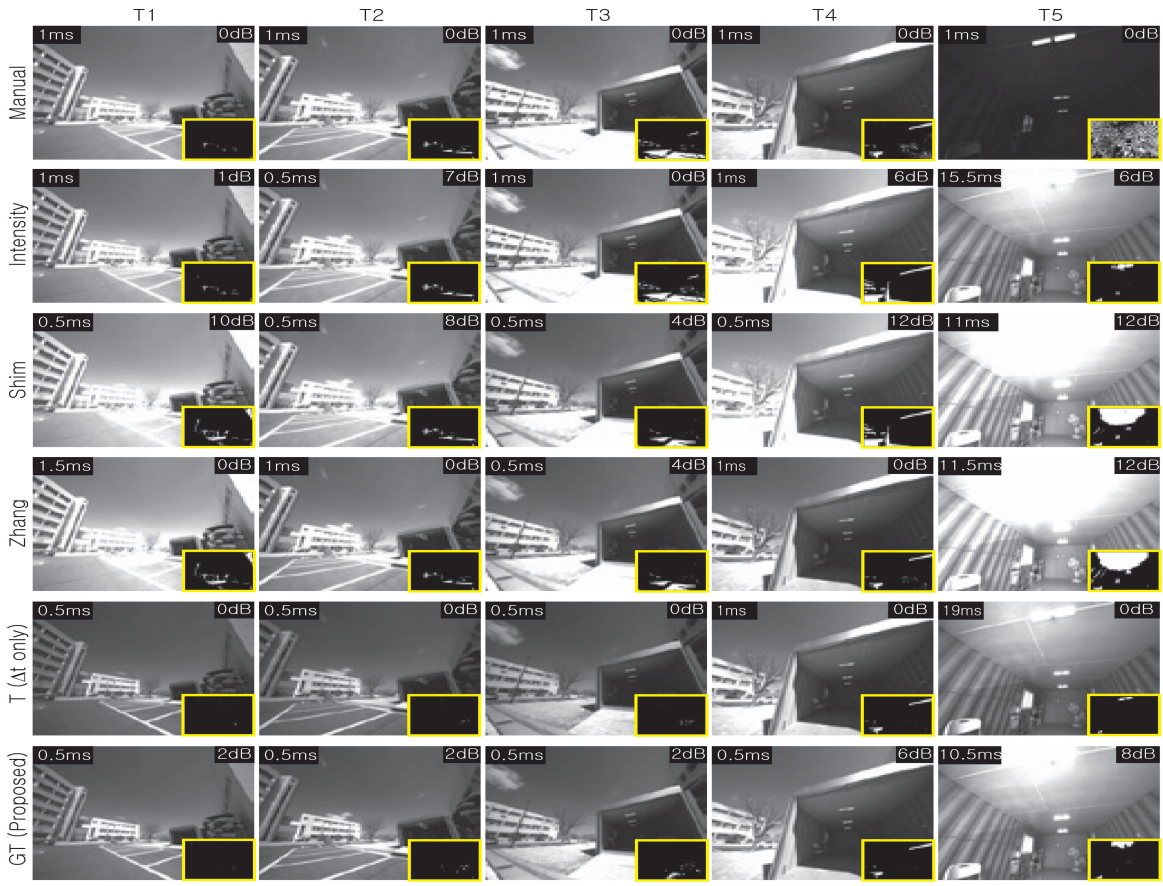


Fig. 10. Each row shows the image captured using optimal attribute determined by each method. Sample optimal images for five time steps are shown (from T1 to T5). The bottom right of the image represents the saturated mask created by the proposed method.

TABLE II
ANALYSIS OF EXPERIMENT FIG. 10

Method	Frame number T1				T2				T3				T4				T5			
	FN	G	%	SNR	FN	G	%	SNR	FN	G	%	SNR	FN	G	%	SNR	FN	G	%	SNR
Manual	397	31.91	0.67	36.31	289	31.49	2.49	36.47	276	26.36	13.65	36.90	168	24.41	3.22	37.20	0	4.75	33.85	39.04
Intensity [14]	412	32.46	0.72	35.91	265	31.83	2.37	35.98	159	26.36	13.65	36.90	136	23.75	11.63	37.18	127	24.45	2.96	36.86
Shim [5]	401	32.62	4.71	34.92	249	31.72	2.56	35.71	192	27.64	6.63	36.22	152	24.63	10.96	36.35	61	26.72	24.49	35.74
Zhang [6]	349	32.00	4.81	35.68	313	31.49	2.00	36.47	176	27.64	6.63	36.22	187	24.41	3.22	37.20	92	26.48	28.44	35.83
T [1]	414	24.00	0.07	38.59	323	26.00	0.11	38.19	188	24.63	0.22	36.81	111	24.41	3.22	37.20	96	21.31	0.98	38.32
GT	449	27.20	0.06	37.65	328	29.48	0.28	37.13	228	27.01	1.60	36.11	218	24.62	2.46	36.94	141	23.95	4.36	36.74

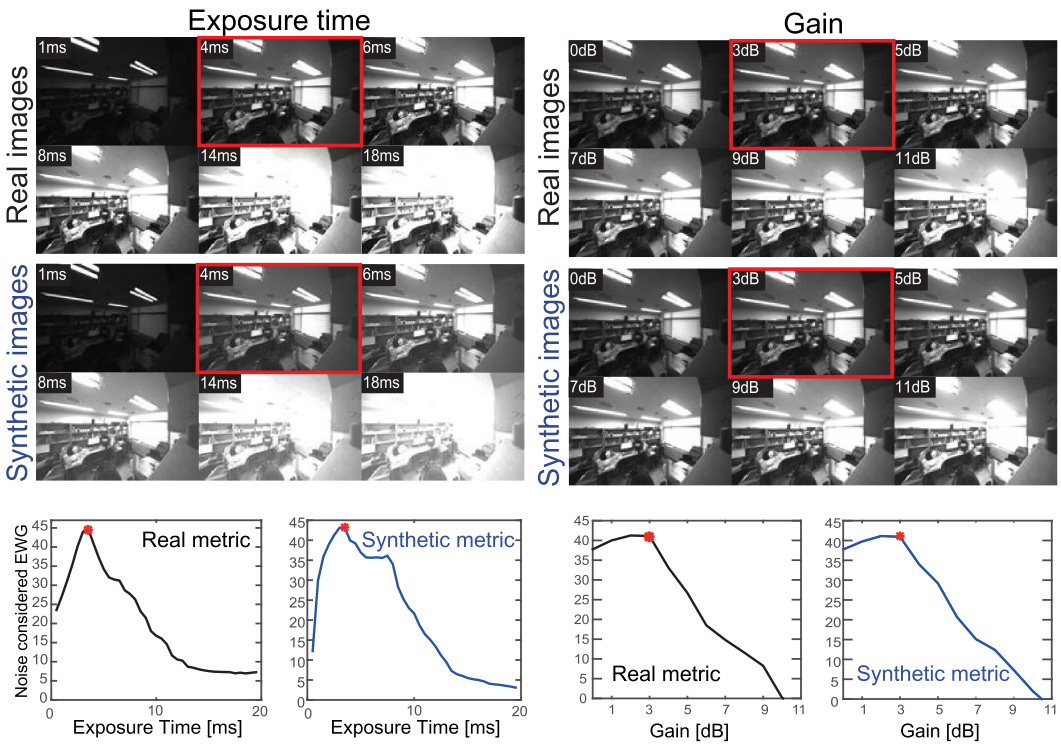
FN means that the tracked feature points are determined by each method, and G is the average of the gradient for the applied exposure time. The saturation rate over the whole image size is written as %.

B. Validation of Image Synthesis

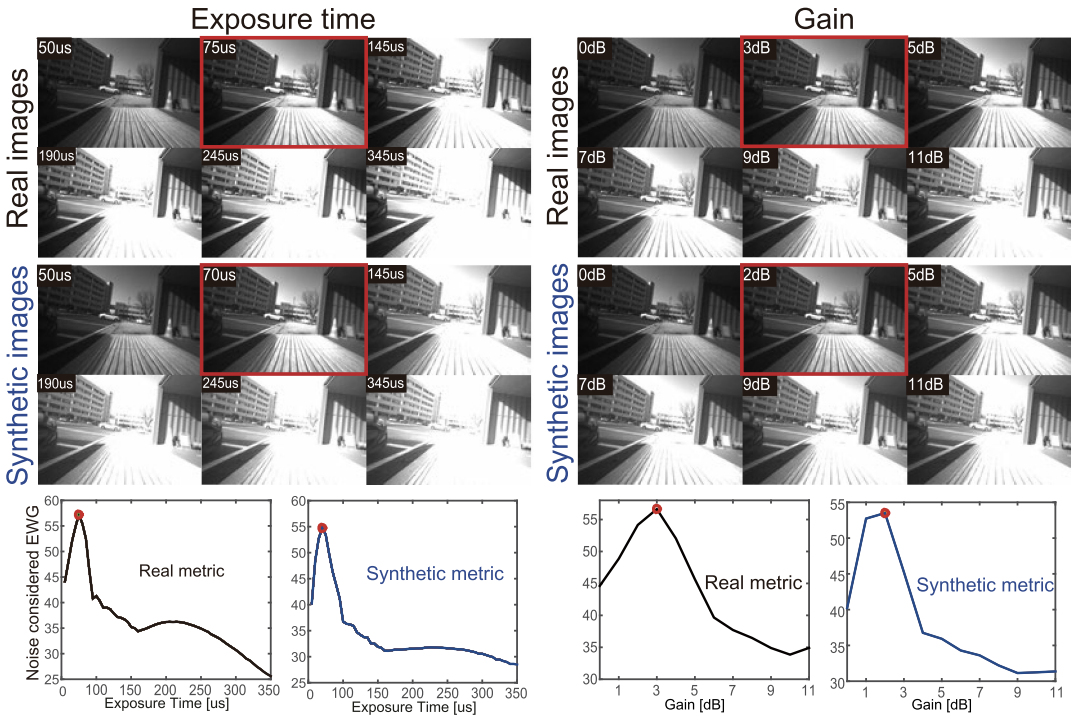
We validated image synthesis in terms of the image quality metric by using the actual images exhaustively acquired while changing the exposure and gain. Images were synthesized for the intended exposure time and gain values [bottom group in Fig. 11(a) and (b)]. For an exposure time-based synthesis, we used a seed image for minimum exposure time to generate other images, and for gain-based synthesis, we used the minimum gain (0 dB) as the seed image.

We would like to emphasize that our intention was not to synthesize images that match real images. Rather, our goal is to perform an image synthesizing process that follows a metric evaluation similar to an actual image. Consistent with this goal, the result of a composite image sequence, as seen in the metric plots of Fig. 11(a) and (b), represented a similar image evaluation metric when compared to the metrics obtained from the actual image sequence. Most importantly, we demonstrated that we can use the proposed synthetic image to find the optimal camera parameters because the chosen optimal value is the same in both the synthetic and the actual image sequences. This image synthesis is included in our control scheme as it is essential for optimal camera property control to perform a fast evaluation. In our previous work [1], we grabbed a real image from the camera to perform image evaluation and control, which took about 5 Hz of total control time. On the other hand, when image evaluation and control enabled us to substantially improve in process time (i.e., 17–46 Hz as evaluated in Table IV).

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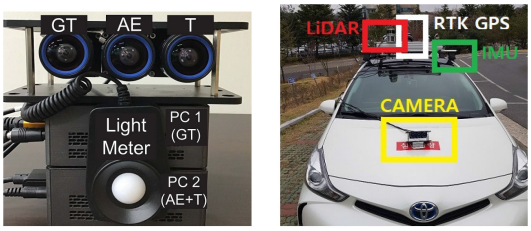
(a)



(b)

Fig. 11. Evaluation of synthetic images. This is a comparison and analysis of synthetic images for camera gain using (14) and synthetic images for exposure time using CRF. Synthetic image rows show how similar they are when synthesized from real seed image of the minimum exposures. Red boxes are the image with the maximum amount of information (NEWG) of the actual and synthetic image. (a) Evaluation of images in an indoor environment. Left column images show an image obtained at 1 ms intervals and synthetic images generated through CRF. Right column images show the image obtained at 1 dB intervals and the synthetic images generated by the gain equation. Bottom graphs represent the metric for the actual image (black) and the synthetic image (blue). (b) Evaluate the image in an outdoor environment. Left column images show the actual image and the synthetic image for the exposure time, and the right column images show the actual image and the synthetic image for the gain. Bottom graphs represent the metric for the actual image (black) and the synthetic image (blue).

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(a) (b)

Fig. 12. (a) Camera rig and a light meter. (b) Vehicle is equipped with a LiDAR, RTK-GPS, IMU, and the camera rig.

C. Camera Attribute Control for Visual SLAM

Lastly, we validated the performance of the proposed camera controller using visual SLAM methods. We performed, in total, three sets of experiments—indoors, outdoors, and a transient environment. For the three sets, we mounted a rig of three cameras operating different control schemes on a mobile robot or a carlike vehicle (see Fig. 12). To evaluate the visual SLAM application, we piped the image sequence from each control method into the ORB SLAM [13] and DSO [4], respectively. See attached multimedia `gcac.mp4`. ORB SLAM leverages binary features, and thus, robust to the illumination changes; DSO utilizes the photometric calibration to handle the illumination variance. We tested using loop-closure modules for ORB SLAM. To incorporate the nondeterministic characteristic, we have run each algorithm ten times and used the median trajectory as our result figure.

DSO provides photometric calibration to mitigate the potential illumination variance. We have used the DSO implementation as in [27] that adopts a photometric calibration in [28]. We evaluated the overall SLAM performance with (Pcalib ON) and without (Pcalib OFF) using photometric calibration. "Pcalib ON" and "OFF" were determined by the availability of photometric calibration following [4]. Specifically, for photometric calibration ("Pcalib ON,") we followed the same procedure in [28], using recorded images and the associated exposure time pairs. For "Pcalib OFF," the algorithm relied on affine brightness transformation.

1) Indoor Corridor at Night: First, the images were captured from corridors at night. As the lighting was automatically controlled by the movement of pedestrians, the lighting was turned ON and OFF repeatedly as the robot moved. We captured the image sequence by hand-holding the camera rig.

In terms of the captured image quality, when the exposure was automatically (`AE`) controlled in an indoor environment, excessive exposure time was allocated for dark scenes. It is noteworthy that at night, when the light is turned ON by the automatic lighting system, the conventional automatic exposure by the global shutter resulted in a large image loss due to erroneous exposure control. Controlling for only the exposure time (`T`) yielded underexposed images in a dark environment. In a dark scene, the auto exposure control (`AE`) assigned a higher gain value to produce a higher noise image and longer exposure time to induce motion blur.

Unfortunately, in the indoor handheld test, no ground truth was available and the monocular SLAM posed scale ambiguity. In the visualization, we corrected the scale by using the corridor length from the blueprint. The trajectory with the median error was selected by comparing the error between the estimated endpoint and the priorly known endpoint from the blueprint. When the illumination changed abruptly [the gray box in Fig. 13(a)], using automatic exposure control caused a feature-tracking failure [bottom right image in Fig. 13(a)]. Both exposure-time-only (`T`) and generic control (`GT`) estimated the trajectory even under a radical light condition change.

In evaluating exposure control with the direct method, we tested with and without using the photometric calibration provided by DSO [4]. Similar to the feature-based method, using automatic exposure caused tracking losses under a sudden light condition change [right image in Fig. 13(b)]. We were interested in comparing the effect of the photometric calibration on-fly compared to our proactive control scheme. As can be seen, the direct method could compensate for over-exposed areas through brightness correction to prevent loss (AE-Pcalib ON), but with drift. As can be seen, the proactive camera attribute control allowed reliable performance in both Pcalib ON and OFF.

2) Outdoor at Sunset: We performed the next set of tests around sunset (3 P.M), facing direct sunlight during the mission. Similar to the indoor case, we evaluated the performances over ORB SLAM and DSO. During the test, the cameras faced the sun in various direction to show different illumination conditions, as shown in the thumbnails in Fig. 14. For the tests, we used a carlike vehicle equipped with VRS-GPS that provided accurate positioning data at 1 Hz. Using these as the baseline, we evaluated the estimated trajectory from each control scheme. In this test, we were able to use the baseline trajectory to correct the scale ambiguity of monocular SLAM. In many field robotics applications, manual exposure in a fixed exposure is often favored in the outdoors. We set automatic exposure (`AE`), manual (`M`), and our (`GT`) control. In the case of feature-based SLAM, proper initialization failed due to the feature points in the sky as also reported by [29], and we could not evaluate ORB SLAM and DSO using the same data sequence. Instead, we performed an additional experiment for feature-based SLAM evaluation in the same environment which is less affected by the sky due to the surrounding buildings.

Compared to the automatic and manual exposure control, the proposed method improved performance and maintained a more consistent trajectory regardless of photometric corrections in direct sunlight in both direct and feature-based SLAM. However, when comparing root-mean-squared error (RMSE) of ten trials, using photometric calibration together with the proposed camera attribute control yielded better performance as shown in Table III.

3) Transient Environment During a Day: We performed the final experiment (see Fig. 15) in an area that incorporated both indoor and outdoor spaces; more specifically, we went from a dark underground parking lot setting to a bright outdoor. Similar to in the outdoor test, we compared the three control schemes by applying them to ORB SLAM and DSO, (i.e., automatic

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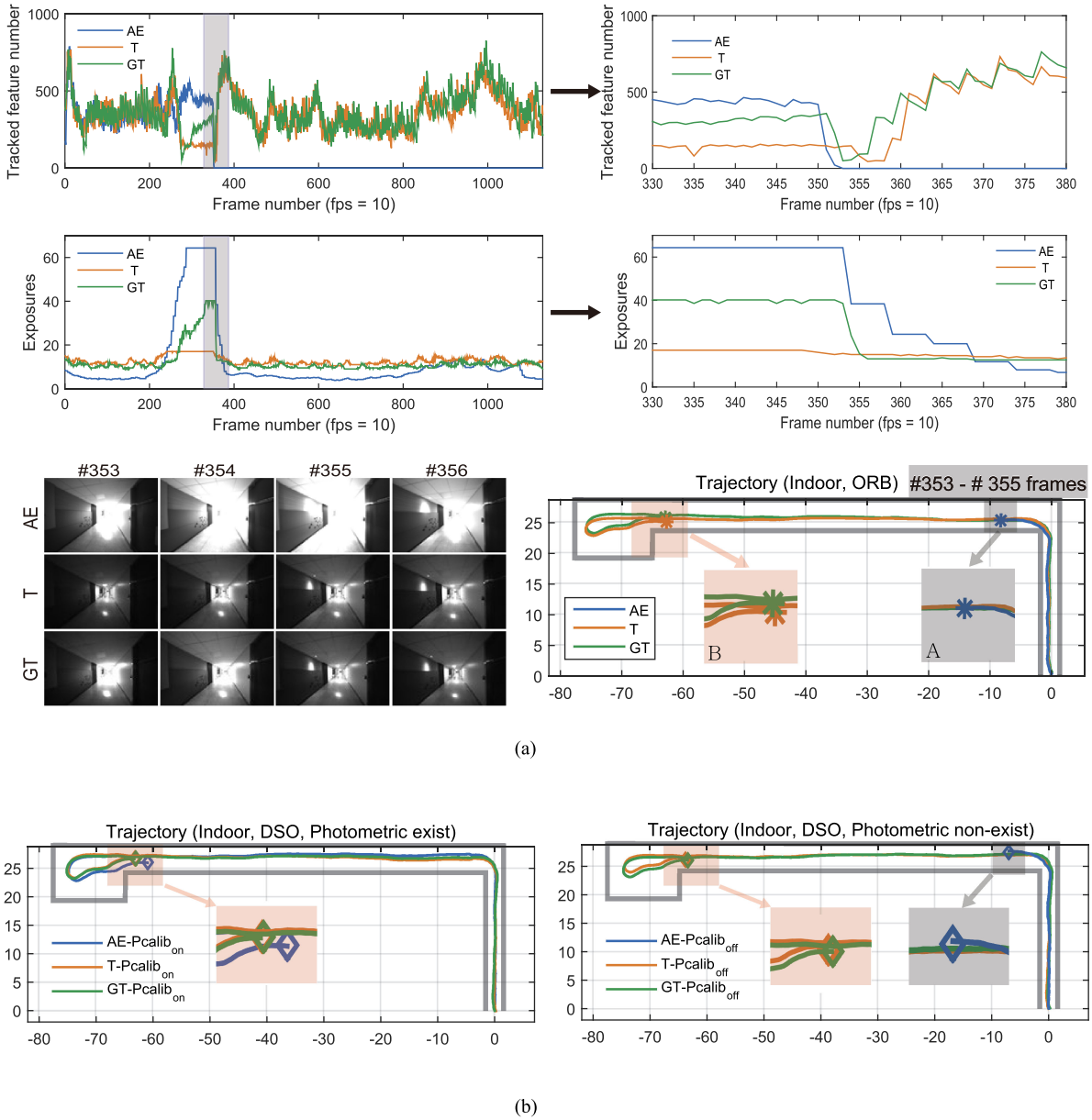


Fig. 13. Indoor corridor experiment at night time using feature-based and direct SLAM. The corridor was operated by an the automatic light system that turned on and off as pedestrians moved, yielding radical illumination changes. The gray box indicates when the light condition changed radically and the red box is the terminating point. AE without photometric calibration and proposed with photometric calibration failed (gray box). (a) Evaluation on feature-based SLAM. The number of tracked feature and corresponding the exposure value with respect to the frame (top left). The graph on the right bottom shows the trajectory estimated. AE failed due to the tracking loss under sudden light change (gray box). (b) Evaluation on direct SLAM. The left shows the DSO estimated trajectory after the photometric correction, and the right shows the nonphotometric correction result.

exposure (AE), manual (M), and our (GT) control). Because we could not guarantee the fixed state of the RTK GPS measurement near a building and indoors, we mounted a LiDAR on the vehicle to leverage two-dimensional (2-D) LiDAR SLAM [26] as the baseline.

Under such a large and radical change in light conditions, no control scheme yielded a reliable trajectory estimation except for the proposed method. The proposed proactive camera attribute control substantially outperformed others in this transient environment given that all other visual odometry failed when the radical illumination change occurred.

Unlike indoor and outdoor tests, radical illumination variance in transient environments resulted in reduced performance of using photometric correction when illumination is changing rapidly [as region "B" in Fig. 15(b)]. Examining the photometrically calibrated images in this region, we found that the photometric calibration may exacerbate the saturation region. This phenomenon was reported by [2], who explained that the CRF curve fitting in the photometric calibration may result in compression of pixel values ranging [0, 90) and stretch of pixel values ranging [190, 255], worsening the image quality. As can be seen from the left image in Fig. 15(b), the drift occurs with

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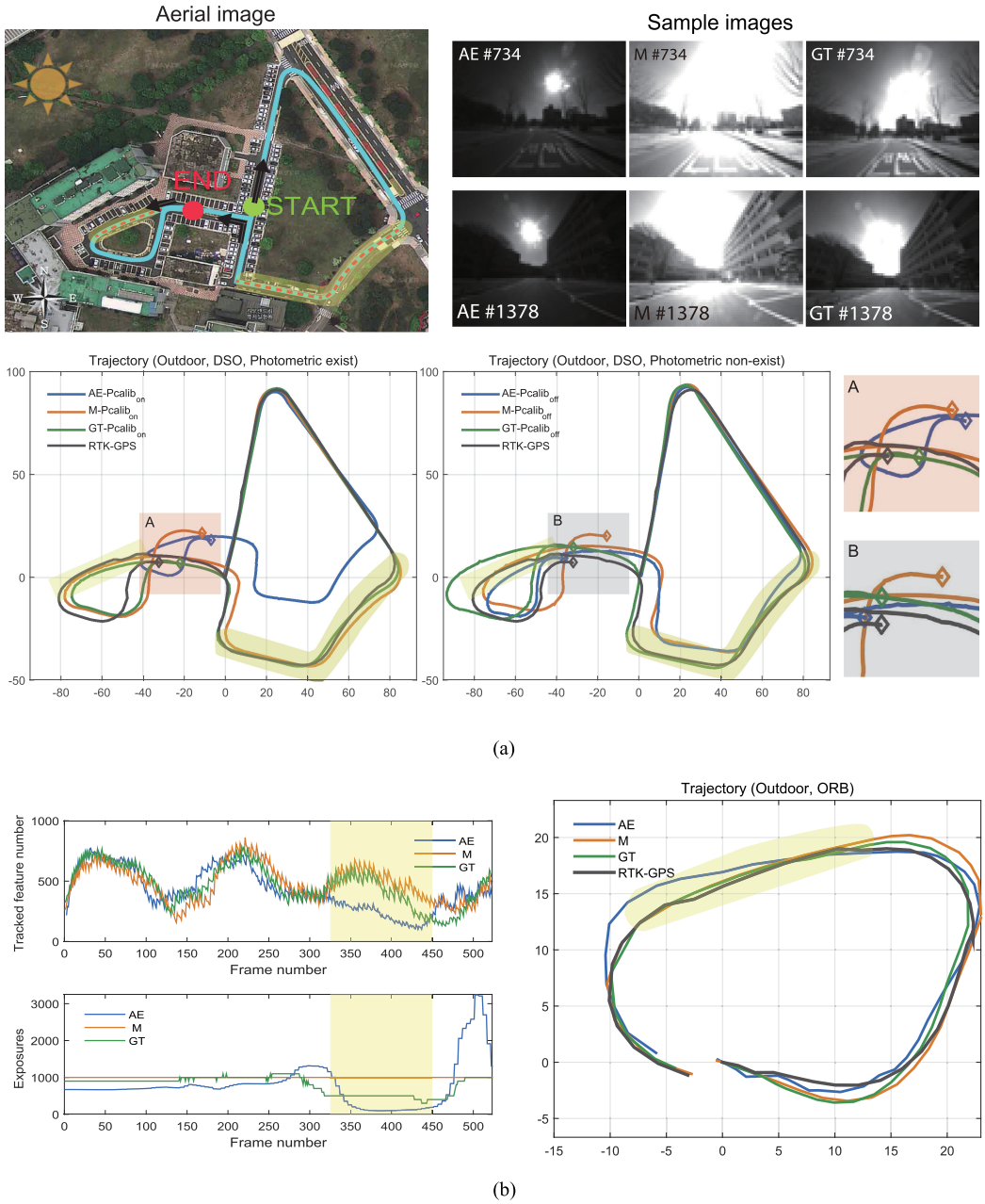


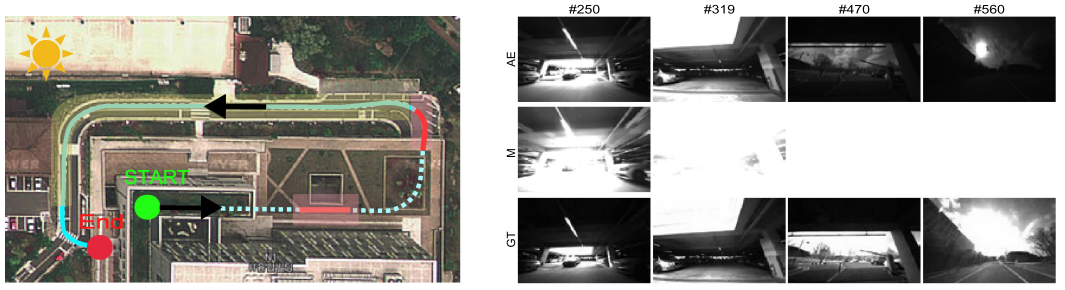
Fig. 14. Outdoor experiment at sunset. The estimated trajectories are compared by using the RTK-GPS position as the ground truth. The highlighted regions (yellow) are where the camera rig faced direct sunlight. The associated sample images are given at the top. The plotted trajectory is without the loop-closure event to show the navigation drift more clearly. In the attached video `gcac.mp4`, the effect of loop-closure is presented. The trajectory is severely deteriorated by the wrong loop-closure from the large illumination variance. (a) Evaluation on direct SLAM. The trajetory is overlaid on a satellite image (top left). The bottom left shows the DSO estimated trajectory after the photometric correction, and the bottom right shows the nonphotometric correction result. (b) Evaluation on feature-based SLAM. The tracked feature points, exposure values, and trajectory results are compared.

the photometric correction due to an induced saturated region from a sudden change in the light conditions.

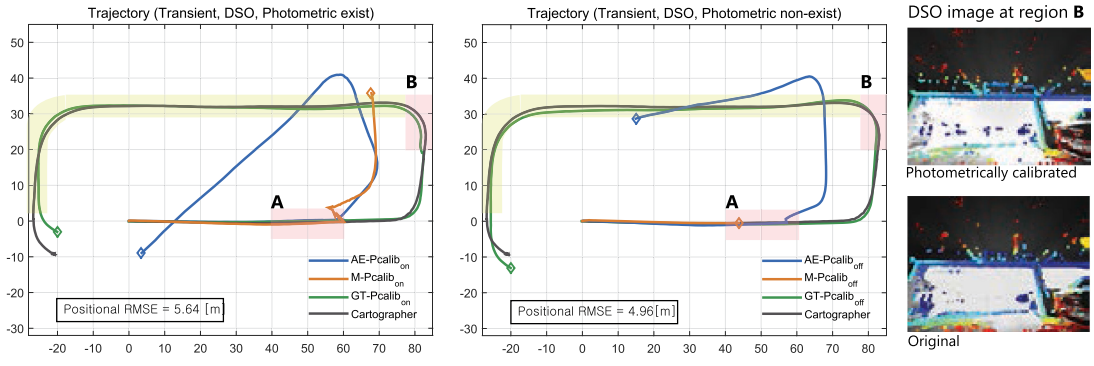
D. Computational Cost

Real-time performance is a key factor in robot vision. Using synthetic images for metric evaluation significantly improved performance and ensured real-time image streams. In this section, we provide a detailed computation time list for each module as summarized in Table IV. We performed the experiments using a mini-PC with Intel Core i7 processors (4 cores@2.40 GHz) and 16 GB of RAM. To simulate the illumination varying condition during the computational time evaluation, we used a controllable artificial LED light to ceaselessly increase and decrease light for the `dynamic` illumination test. The computational times are reported according to: 1) when we selectively performed BO based on the global illumination change (`selective`); 2) when we applied the entire control scheme in a greedy manner (`batch`) by computing attributes for every image.

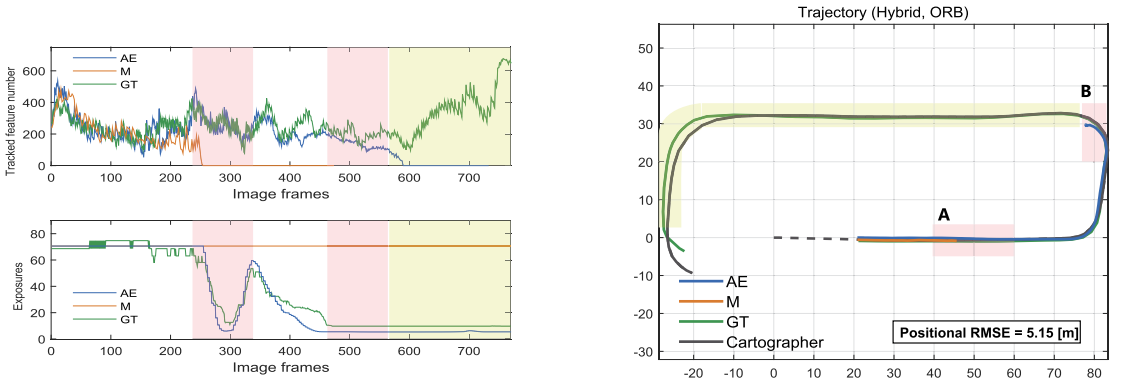
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(a)



(b)



(c)

Fig. 15. ORB SLAM and DSO evaluation in a transient environment. (a) Indoor region (cyan dotted line) and outdoor region (cyan solid line) are overlaid on a satellite image together with the vehicle path. The red highlighted box indicates where the sudden illumination change occurs and the yellow highlighted box represents the region facing the direct sunlight. On the top right are the sample images from the automatic exposure (AE), manual (M), and our (GT) control. (b) and (c) Both direct and feature-based are compared against the baseline trajectory generated by 2-D LiDAR SLAM [26].

TABLE III
WE COMPUTED POSITIONAL RMSE OF THE TRAJECTORY BY COMPARING THE BASELINE AGAINST RTK-GPS

Outdoor 1	ORB						
Ctrl mode	AE		M		GT		Path length [m]
RMSE [m]	2.62		2.58		2.45		97
Outdoor 2	DSO						
Ctrl mode	AE		M		GT		Path length [m]
Photo calib	ON	OFF	ON	OFF	ON	OFF	
RMSE [m]	28.71	8.52	10.54	9.88	5.60	7.57	515

ON and OFF indicate the photometric calibration option in DSO.

TABLE IV
TIME REQUIRED FOR EACH MODULE

Global Illum. Check	Illum. Cond.	Img. Synth. [us]	Metric eval. [us]	Total Control [us]	[Hz]
Batch	Static	196	3273	58215	17.27
Batch	Dynamic	185	3210	58484	17.22
Selective	Static	207	4339	21385	46.79
Selective	Dynamic	205	3396	23553	44.33

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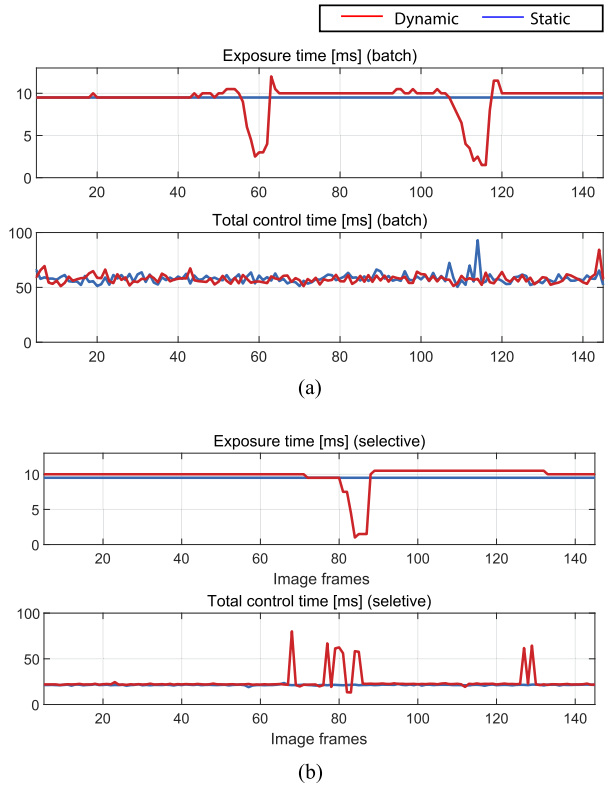


Fig. 16. (a) Exposure time changes with respect to the illumination change in dynamic illumination (red) regardless of batch and selective control mode. Using batch mode shows similar overall control time for static and dynamic illumination. (b) Applying selective mode is also capable of adjusting exposure control adequately while the overall control time only increased to request BO only when light condition changes.

As can be seen from Table IV and Fig. 16, checking global illumination change and selectively performing BO improved the speed substantially. The performance was reliable regardless of the illumination condition if the lighting is constant or changing. Although we found the `selective` mode is sufficient with a higher frame rate, the `batch` mode would be preferred for large illumination variance with a lower frame rate. One example would be underwater operation such as [30], that experiences large illumination variance between the water surface and underwater with a camera running at 1–2 Hz.

VIII. Conclusion

In this article, we proposed a proactive and generic camera attribute control scheme using BO. Simultaneously controlling exposure time and gain using fast functional evaluation through synthetic images without acquiring actual images with CRF-based image synthesis increased learning efficiency. We conducted extensive evaluations in terms of image quality metrics, feature tracking, and overall SLAM performance. The proposed method substantially improved the image acquisition, reducing tracking loss even under a severe illumination change for the feature-based method. The proposed camera attribute control is also beneficial to the direct SLAM methods. Although the proposed method does not explicitly impose photometric consistency between frames, pursuing a more informative image even under radical illumination variance reduced saturation-enriching image details. We believe this proactive camera control would be particularly beneficial for a transient environment with a large illumination discrepancy (e.g., water surface to underwater, indoor to outdoor, and day to night).

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